POLLUTION HAVENS AND THE TRADE IN TOXIC CHEMICALS:

EVIDENCE FROM U.S. TRADE FLOWS

by

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Abstract

Does increased environmental protection decrease the emission of pollutants or merely displace them? Using newly available trade data, this study examines the flows of a panel of chemicals designated as toxic by the U.S. Environmental Protection Agency's Toxics Release Inventory (TRI). Estimates from a differences-in-differences model indicate a significant increase in net imports when a chemical is listed on TRI, which suggests production offshoring. Furthermore, I find that increased imports due to this "pollution haven effect" are sourced disproportionately from poorer countries, which are likely to have lower environmental protection standards. At the same time, I observe the bulk of American trade in toxic chemicals occurs with other wealthy countries, which may be attributed to the capital intensity of chemical production.

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1. Introduction

Do changes in environmental regulation affect where pollution-intensive goods are produced? It has been argued that variations in environmental policy lead to the emergence of "pollution havens," with polluting activity relocating to areas with less stringent regulation.[1] This issue, often raised amid international trade negotiations, is increasingly relevant with rising globalization and heightened awareness of environmental hazards.

This paper examines United States trade in toxic chemicals. In particular, I explore the pattern of trade in chemicals designated as toxic by the U.S. Environmental Protection Agency's Toxics Release Inventory (TRI) program. Following the deadly industrial accident at a Union Carbide plant in Bhopal, India and a chemical release at a similar plant in West Virginia, the Emergency Planning and Community Right-to-Know Act (EPCRA) was signed into law in 1986.[2] The act required domestic industrial facilities to report to TRI the quantity of releases and transfers of certain toxic chemicals. These data are made available to the public under the premise that this information creates incentives for companies to improve their chemical management and reduce toxic releases. TRI data collection began in 1988 with 332 chemicals listed as toxic and has increased coverage to the current 666.

This study focuses on two basic questions. First, I examine whether the TRI program has resulted in the relocation of industrial activity, which is sometimes called the pollution haven effect (PHE). If TRI increases the costs associated with producing, using, and processing a particular chemical, then for a given level of domestic use, one might expect domestic production to be replaced by an increase in net imports.[3] Second, I examine where these imports originate. A popularly held belief is that pollution-intensive industrial activity relocates to lower-income countries, which tend to have less-stringent environmental policies. This phenomenon, called the pollution haven hypothesis (PHH), is controversial in part because it is not obvious that poorer countries, given their factor endowments, would have a comparative advantage in producing capital-intensive goods like chemicals and thus affect trade patterns.[4]

To analyze the trade of TRI chemicals over the past two decades, I use newly available trade transaction data from the import and export records collected by U.S. Customs and Border Protection and processed by the Foreign Trade Division of the U.S. Census Bureau. These records comprise the universe of commodity shipments that enter and leave the U.S. every year since 1992, totaling over 900 million unique transactions. Each transaction record provides highly detailed information, such as commodity category, quantity in dollars and physical units, transaction date, country (or U.S. state) of origin and destination, ports of entry and exit, mode of transport (land, sea, or air), whether it is a related party transaction, etc. Besides not having been used before to explore the relationship

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[1] See Jaffe et al (1995), Copeland and Taylor (2004), Brunnermeier and Levinson (2004), and Taylor (2004) for surveys.

[2] Note that the jurisdiction of EPCRA covers only the United States; however, like with labor practices, media coverage and public activism may lead American multinational firms to extend domestic (or at least improve) standards to their foreign operations. This is consistent with the differential impact of related party trade percentages depending on whether the chemicals were imported or exported; see Tables 2 and 3.

[3] One might also expect this effect to be stronger for chemicals that are more toxic and/or are released into the environment in greater quantities (and therefore are more noticeable to the public). This area I intend to explore in future research.

[4] Confusingly, the pollution haven *effect* has sometimes also been called the pollution haven *hypothesis*; in this paper, I maintain the distinction between the two. A third related concept is the factor endowment hypothesis (FEH); see text for further discussion of these issues.

between trade and the environment, these data also provide a methodological advantage. Unlike previous literature in this area, I am able to observe trade at the commodity-level, instead of the industry- or subsector-level, and my measure of regulation is also at the commodity-level. This mitigates concerns regarding the composition of a subsector's output, some of which may be pollution-intensive and subject to regulation while others are not.

For my analysis, I use a differences-in-differences model to identify a discrete change in net imports before and after a chemical is listed on TRI while controlling for factors and trends also found in reference groups of unaffected chemicals.[5] This identification strategy is further sharpened by the variation within my panel of chemicals, which differ in their dates of TRI listing or delisting; trade volumes; chemical properties and usage; and countries of origin and destination. I also control for other factors that may influence trade in general, such as tariff rates, bilateral distance, foreign direct investment, and trade between related parties.

Estimates from the model indicate that TRI listing corresponds to a marked increase in the net imports of the affected chemicals; concurrent with a decrease in domestic emission levels, this clearly suggests production offshoring due to regulation, i.e., a pollution haven effect. I also fail to reject the pollution haven hypothesis since the increase in net imports appears to be disproportionately sourced from countries with lower per capita income, which I use to proxy for environmental stringency. Interestingly, these findings are still consistent with the factor endowment hypothesis since much of the absolute volume of trade for the panel of chemicals occurs with other wealthy countries, which presumably have a comparative advantage in producing capital intensive goods.

Thus, while it may appear that TRI is having its intended regulatory effect of encouraging facilities to reduce their toxic releases, it may be that only the nationality of these emissions has changed, not their net environmental impact. Furthermore, these imports may offset, even dominate, domestic factors like improvements in emission capture, substitution with less toxic substances, and lower market demand. Perhaps most ironic, however, is that given the relocation of production to countries with lower environmental standards, it is possible that the worldwide total of emissions for these chemicals has increased due to domestic regulation.

The paper proceeds as follows. In Section 2, I discuss some of the prior literature on trade and the environment, with a particular focus on studies related to the idea of pollution havens, while Section 3 presents the TRI program in greater detail. In Sections 4 and 5, I describe the data used in the analysis and my methodological approach, respectively. I then discuss the findings in Section 6 and offer some concluding remarks in Section 7.

2. Previous Literature

Although there is a growing literature examining the impact of trade (and freer trade) on the environment, from a theoretical standpoint the relationship between the two is ambiguous.[6] First, there is a simple scale effect.[7] All else being equal, if trade spurs an increase in economic activity, then it unambiguously leads to an increase in pollution. Second, there is a composition effect. That is, trade alters the composition of industrial activity in

[5] These reference groups include balanced and unbalanced sets of chemicals that are not in the panel of "treated" chemicals (i.e., those experiencing a change in their TRI-listing status); see Section 5 for details.
[6] A more formal theoretical treatment appears in Antweiler et al (2001).
[7] This tri-partite framework is from Grossman and Krueger (1993), who in turn claim to have borrowed it from a report by the Commission of the European Communities from a few years earlier.

nations, according to their comparative advantage. Some industries are obviously more polluting than others. Whether pollution decreases worldwide as a result of freer trade is ambiguous, depending on whether polluting activity shifts towards or away from countries with greater environmental regulation, which in turn may depend on whether traditional sources of comparative advantage (e.g., factor endowments) outweigh advantages of laxer environmental regulation. Finally, there is a technique effect. Specifically, the stringency of a nation's environmental regulation and the pollution-intensity of its industrial activity may change following trade liberalization. For instance, if environmental quality is a normal good, then as trade increases income, the public's demand for more stringent pollution controls will increase. Foreign firms opening facilities in less developed countries may also bring with them better pollution abatement technologies.

The empirical question is whether the technique and composition effects outweigh the composition and scale effects, recognizing that the composition effect is itself ambiguous. In their study of the possible effects of NAFTA on Mexico and the United States, Grossman and Krueger (1993) find that Mexico at the time was exactly at a level of per capita income where further increases in income would likely lead to increased demand for environmental protection. In addition, they find that freer trade would likely result in Mexico further specializing in industries employing low-skilled workers (its comparative advantage), which tend to be less pollution-intensive. On balance, therefore, they concluded that NAFTA might well improve the Mexican environment, counter to the concerns of environmental groups at the time.[8] Antweiler et al (2001) model and estimate the scale, composition, and technique effects of international trade on sulfur dioxide (SO_2) concentrations and find that the composition effect is negative (but small) for the average country, and the scale and technique effects are (together) negative. Their findings suggest that free trade is indeed good for the environment. Looking at seven different pollutants and their environmental outcomes, Frankel and Rose (2005) find that trade is beneficial to the environment on a few measures, has no effect on others, and potentially has a detrimental effect on one (carbon dioxide).

Embedded within this larger scale/composition/technique structure are two separate notions that have received significant attention and debate. One, taken together, the scale and technique effects trace out what is now called the environmental Kuznets curve (EKC), an inverse-U relating per capita income and pollution. Launched by Grossman and Krueger (1993), there are a large and growing number of papers estimating this relationship (and the top of the hump) for a varying set of pollutants, geographic regions, and time periods. The second set of literature deals with the ambiguous composition effect, and in particular, which countries attract polluting industries when trade is liberalized. This paper addresses this latter phenomenon.

On this issue, the literature has identified three main hypotheses. One, dubbed the pollution haven effect (PHE) by Copeland and Taylor (2004) in their survey of the subject, states that an increase in a country's regulatory stringency will, at the margin, result in a relocation of pollution-intensive industrial activity and a related impact on trade flows. This notion has strong theoretical support, but early empirical studies essentially found no evidence of PHE – that is, that environmental regulation affects the (domestic or international) location of industrial activity.[9] On the specific question of international trade, studies have examined whether, all else equal, a country's net exports

[8] However, if Mexico were also to become more capital-abundant – for example, from more foreign direct investment – then the impact is more ambiguous.
[9] See Jaffe et al (1995) for a review of this literature.

(or imports) of an industry's products is dependent on the domestic regulation faced by the industry, often as measured by the industry's relative pollution abatement costs.[10] These studies have generally found no significant relationship.

However, more recent studies have overcome the empirical issues that plagued the earlier research and do in fact find significant "pollution haven effects." Some of these studies focus on regulatory differences and industrial relocation within the United States.[11] Others examining the impact on foreign trade find an increase in net imports due to domestic regulation, as proxied by pollution abatement costs.[12]

A second hypothesis related to the composition effect, and a popularly held belief, is the pollution haven hypothesis (PHH). It states that trade liberalization will result in a shift of pollution-intensive industrial activity from countries with stringent environmental regulations to those with weaker regulations. In a common variant of this hypothesis, the shift is from high-income countries to low-income countries, which assumes that environmental quality is a normal good and therefore environmental regulation increases with per capita income. A corollary to this hypothesis is that pollution decreases in high-income countries and increases in low-income countries. Because they are closely related, and the distinction between them is often blurred, it is worth noting that the PHE is necessary (but not sufficient) for the PHH to be true.

The third hypothesis, and the chief alternative to the PHH, is the factor endowment hypothesis (FEH). It states that trade liberalization will result in a shift of pollution-intensive industrial activity toward countries relatively abundant in the factors used in such activities. To a large extent, pollution-intensive industries are capital intensive and therefore a shift to capital-abundant countries would be predicted. What is noteworthy is that capital-abundant countries tend to be high-income countries with stricter environmental regulations. Therefore an interesting possibility is that the PHH and the FEH are both valid but they work in opposite directions of each other and the resulting shift in trade patterns would depend on which effect is stronger.[13] Moreover, if pollution-intensive activity shifts towards more regulated countries, it is possible for global pollution to decrease, although these effects are likely to differ by industry, pollutant, or country.

In terms of evidence for the PHH, Low and Yeats (1992) is an oft-cited study showing that the share of pollution-intensive goods exported by poor, developing countries increased between 1966 and 1985. In another well-cited study, Hettige et al (1992) similarly find that the toxic intensity of industrial production grew most quickly in the 1970s and 1980s for lower-income countries. While consistent with the PHH, Copeland and Taylor (2004) describe how these trends might also be consistent with the PHE alone, or with the FEH (e.g., increasing capital abundance in lower-income countries during this period). That most pollution-intensive production still occurs in high-income – and typically heavily regulated – countries clearly suggests that factors besides environmental policy also play a role.

A number of recent studies suggest this as well. Antweiler et al (2001) find support for the FEH, and in

[10] Kalt (1988); Tobey (1990); Grossman and Krueger (1993).
[11] E.g., the county-level focus of Becker and Henderson (2000).
[12] E.g., Ederington and Minier (2003) and Levinson and Taylor (2008). Brunnermeier and Levinson (2004) and Copeland and Taylor (2004) offer reviews and discussions of both waves of this literature.
[13] Antweiler et al. (2001).

particular that high-income, capital-abundant countries have a comparative advantage in pollution-intensive, capital-intensive products. However, the composition effect appears to be small, suggesting that the effect predicted by the PHH is largely negating. Cole and Elliott (2003) also find evidence of both the FEH and the PHH canceling each other out, for the air pollutant emissions of SO_2 and somewhat for carbon dioxide (CO_2). Similar results, however, are not found with nitrogen oxides (NO_x) and biochemical oxygen demand (BOD), a measure of water pollution. Frankel and Rose (2005) find no evidence for either the PHH or FEH. Meanwhile, Ederington et al (2004) find that the U.S. manufacturing sector shifted toward less pollution-intensive industries between 1972 and 1994, although this shift also appears in U.S. imports. They find that this manufacturing shift is not due to trade liberalization; in fact, tariff reductions may have shifted U.S. industrial production *towards* more pollution-intensive industries. Furthermore, they find no evidence for the PHH, in that tariff reductions had a smaller effect on pollution-intensive imports from non-OECD countries than OECD countries.

In examining the relationship between environmental impact and trade, however, much of the focus has been on how changes in trade policy affect environmental outcomes. Less studied, but no less important, is the converse—how changes in environmental policy affect trade in commodities subject to such regulation. That is, given the interest of producers to maximize profits, avoidance of costs associated with regulatory compliance or possible legal liability is likely to incite changes in production technique or relocation to less stringently regulated locales. If the latter occurs, then the consequences may be manifest in changing trade patterns and the origins and destinations of such trade. More to the point, changes in environmental as opposed to trade policy, ceteris paribus, can have both the pollution haven effect and pollution haven hypothesis as possible outcomes.[14] This paper's research design exploits such regulatory change as embodied in the Toxics Release Inventory (TRI) program and the recently available trade transaction data, which are both described below.

3. The Toxics Release Inventory

On 3 December 1984, over 40 tons of methyl isocyanate gas leaked from a Union Carbide plant in Bhopal, India, killing thousands.[15] In January 1985, the EPA announced that 28 leaks of methyl isocyanate had occurred in the previous five years at a similar Union Carbide plant in Institute, West Virginia, and on 11 August 1985, that same plant experienced a chemical release that hospitalized 135 workers and nearby residents. These events are often cited as being among the primary motivations for the federal Emergency Planning and Community Right-to-Know Act (EPCRA), which was signed into law in October 1986.

EPCRA instituted four new reporting requirements for facilities within the United States that use, produce, store, or release certain specified chemicals. Among the four, EPCRA Section 313 requires affected facilities to submit to the EPA and the relevant state office, by 1 July each year, information on releases into the environment

[14] Thanks to Arik Levinson for pointing out the applicability of pollution haven theories to regulatory as opposed to trade policy changes.

[15] The number of deaths remains uncertain to this day. One commonly cited figure is that at least 3,800 died immediately, and it is commonly believed that subsequent deaths bring the total to a few times that initial amount, with 100,000 or more permanently injured.

and off-site transfers of certain toxic chemicals.[16] The information collected by "Form R" is compiled and made available to the public in the Toxics Release Inventory (TRI). The TRI program and the chemicals covered by it are the focus in this study. The first TRI data were released in June 1989, for reference year 1987.

Facilities are required by law to report to TRI their toxic releases and transfers if the facility meets certain conditions. First, the facility must have 10 or more full-time equivalent employees. Second, the facility must be classified as a manufacturing facility (SIC 20-39) or be engaged in one of a number of non-manufacturing activities.[17] Executive Order 12865, signed in August 1993, extended reporting requirements to federal facilities, regardless of their SIC industrial classification. Finally, the facility must manufacture, import, process, or otherwise use any of the listed chemicals in amounts greater than the threshold quantities. For most of the chemicals covered by TRI, the critical annual thresholds are 25,000 pounds for the manufacture, import, or processing of a chemical, and 10,000 pounds for the mere use of a chemical.[18] Over the first 20 years, the number of facilities reporting to the TRI has ranged from approximately 22,000 to 25,400 per year.

At TRI's inception, 332 chemicals and chemical categories were subject to reporting.[19] Through the years, chemicals have been added to the list and others deleted from the list. Currently, for reporting year 2008, there are 666 chemical entities subject to TRI reporting. In addition to some basic information about the facility itself, for each of these chemicals meeting any of the above thresholds, the facility is required to report: how the chemical is used at the facility; the maximum amount of the chemical at the facility at any one time during the year; releases (in pounds) of the chemical to the environment during the year, classified by on-site media (air, water bodies, underground, and land); transfers (in pounds) of the chemical to off-site locations during the year (e.g., disposal facilities, or treatment and recycling facilities); on-site waste treatment methods related to the chemical and their efficiency; source reduction and recycling activities related to the chemical. In 2008, there were 21,943 facilities submitting 84,246 chemical-level reports.[20]

The facility-level data collected by the TRI program (and tabulations thereof) are made available to the public through a variety of means and by a variety of entities. Previously this meant paper publications and CD-ROMs. These days, the internet lends itself particularly well to this sort of data dissemination. The chief source for this

[16] Meanwhile, EPCRA Section 302 ("Emergency Planning") requires facilities to notify local officials of the presence of any of the 356 extremely hazardous substances (EHS) if it exceeds the "threshold planning quantity." EPCRA Section 304 ("Emergency Release") requires facilities to notify local officials in the event of a release of one of the specified hazardous chemicals (EHS plus those defined under CERCLA) in excess the "reportable quantity." EPCRA Sections 311-312 requires facilities to notify local officials if it has critical quantities of any hazardous chemicals, as defined by the Occupational Safety and Health Act (OSHA).

[17] Non-manufacturing facilities were exempt until reference year 1998. The non-manufacturing industries currently in scope to the TRI include SIC 10 (except 1011, 1081, and 1094); SIC 12 (except 1241); coal- and/or oil-combusting facilities in SIC 4911, 4931, and 4939; "RCRA facilities" in SIC 4953; SIC 5169; SIC 5171; and solvent recovery facilities in SIC 7389.

[18] Effective reporting year 2000, the thresholds for certain chemicals were lowered to 100 pounds per year, and for others that are particularly persistent in the environment and highly bioaccumulative, the threshold was set at 10 pounds per year. For dioxin and dioxin-like compounds, thresholds were set at 0.1 grams per year.

[19] The number of chemicals reported may not correspond to the number of chemical-level reports as the latter includes groups of chemicals (e.g., zinc compounds).

[20] Some numbers like pounds of emissions may be subject to revision depending on whether reporting errors are corrected, and other numbers (e.g., facility count) depend on inclusion of various subsets/types.

information is EPA's TRI website[21], but alternative interfaces to search and compile the same data have been created, such as the National Institutes of Health's National Library of Medicine's TOXNET and TOXMAP, OMBWatch's Right-to-Know Network, Environmental Defense's Scorecard, and MapEcos (supported by Duke, Harvard, and Dartmouth). By far, the most widely used and cited items from the TRI are the total on-site releases, total off-site transfers, and the sum of the two, classified by facility, chemical, industry, and/or geography.

The basic premise behind TRI is that making this information publicly available creates incentives for companies to improve their chemical management and reduce toxic releases. The mechanism that is chiefly cited here is community pressure on local facilities; however, consumers and investors can obviously influence companies through market signals. Meanwhile, environmental groups (and journalists) play a role in amplifying the information released by TRI to get companies, local communities, consumers, investors, legislators, and regulators to take action. In any event, TRI is often held up as a successful market-based solution and alternative to the usual command-and-control regulation: Between 1988 and 2008, pounds of on-site toxic disposal and releases into the environment have decreased 41 percent, while the total amount of disposal and releases has decreased 45 percent.[22]

While evidence from facility reports to EPA suggests that the program is achieving its aim of reducing toxic chemical emissions following their listing to the inventory, it is problematic to use these figures to attribute causal impact. This is because facilities are not required to report emissions when the chemicals are not listed, making it impossible to compare pre- and post-listing trends based on emissions alone. This problem of identification is exacerbated by the non-disclosure of firm production data to the public (and the EPA provides only broad ranges of values), as it is not obvious that changes in emission levels owe to changes in levels of domestic production, improved emissions capture, false reporting, gaming of reporting thresholds, overlapping regulation, or a combination of these. A number of studies have found that facilities under-report their toxic releases or engage in "threshold regarding," which would exempt them from reporting their releases.[23] Other studies have suggested that declines might be due to regulation of some of these chemicals under other federal and state programs and/or might be a side-benefit of the regulation and abatement of (non-toxic) pollutants such as those covered under the Clean Air Act.[24]

Although some studies have suggested that the benefits of TRI have been overstated, others suggest that it has in fact had an impact. The economics literature has focused on the role of investors, with various event studies. Hamilton (1995) finds that publicly-traded firms reporting to TRI experienced statistically significant abnormal negative stock returns on the day following the release of the very first TRI data in June 1989. The effect was more pronounced for companies reporting a greater number of TRI chemicals, and less pronounced for those with known Superfund sites. The study also examines the characteristics of companies that significantly impacted whether they received print media coverage in 1989 regarding their TRI emissions. Konar and Cohen (1997) find that, on

[21] http://www.epa.gov/tri/

[22] These calculations do not adjust for the relatively toxicity of a pound of each of these chemicals. It is also worth noting that data for 1987 (the first year of TRI) is routinely dismissed as unreliable. From early on, EPA measured changes using 1988 as the base year.

[23] E.g., Koehler and Spengler (2007), which also provides a nice summary of studies on under-reporting, and Bennear (2005) on threshold regarding.

[24] Koehler and Spengler (2007) and Bui (2005).

average, a publicly-traded firm reporting to TRI experienced a statistically significant abnormal negative stock return on the day following the first print media mention of the company's TRI emissions in 1989. Those companies with the largest drop in stock price then experienced greater declines in their TRI emissions than their industry peers, suggesting the presence of market incentives. Khanna et al (1998) find negative stock market returns for chemical companies in the days following the release of new TRI data, over a series of years. In turn, their stock market losses resulted in reductions in on-site toxic releases, but no change in total toxic waste generated (i.e., on-site releases plus off-site transfers). However, this too suggests the presence of market incentives, as off-site transfers is likely to be more socially responsible than on-site releases into the environment.

Investors may be reacting to various possibilities. TRI emissions may suggest future liability, future regulation, and future pollution abatement costs. Relatively high TRI emissions might also indicate inefficiencies in the production process, a general lack of innovation, and/or poor management. Adverse publicity may affect consumer behavior (e.g., boycotts). And TRI data are often considered by investors and mutual funds interested in Socially Responsible Investing (SRI). In turn, firms may be motivated by declines in share price because it may impact the future cost of capital, managers' own stakes in the company (and potentially their jobs), takeover threats, and simple corporate pride.

This paper continues this line of research on the impact of TRI on production, focusing on two basic questions. First, I look for evidence of whether the TRI program has caused a pollution haven effect (PHE). The intent behind TRI is to create incentives for companies to improve their chemical management and reduce toxic releases, and aggregate evidence suggests that the TRI program has had its intended effect in lowering reported toxic emissions over time. I assert that this decrease in domestic emissions is not wholly benign, however, and that international trade may potentially be a factor in this. For example, if domestic use remains the same, domestic production can be replaced by an increase in net import volumes.[25] This would be potential evidence of industrial relocation and, hence, a pollution haven effect.

Second, I look for evidence that may support the pollution haven hypothesis (PHH). In particular, I examine whether the levels and rates of change in the net imports of the panel of toxic chemicals show different and persistent trends following TRI listing depending on the exporting country's per capita income. Assuming that environmental quality is a normal good, wealthier nations are likely to have stricter environmental policies, and thus a disproportionate share of toxic chemical production should relocate to poorer countries, which may subsequently export production that was displaced by domestic regulation.[26]

Both questions are amenable to a differences-in-differences approach, where I compare changes to net import values for specific chemicals both prior and following TRI listing or delisting. I focus on a panel of toxic chemicals

[25] Since domestic production figures are not disclosed to the public by firms or regulators, I rely on emissions to approximate production. This may be a strong assumption, however, as it ignores improvements in emissions capture or substitution to alternative chemicals. Nevertheless, since TRI-listed chemicals are designated as such due to their known toxicity and emissions are reported for facilities that produce or use these chemicals, changes in the amount imported arguably raises the same concerns about potential exposure as those regarding domestic production and/or use.

[26] A more rigorous analysis, which I intend to perform in future work, would compare net imports from countries that have similar environmental policies, such as TRI-like pollution release and transfer registers (PRTRs), particularly if they cover the same chemical.

that were either added or deleted from the TRI list between the years 1992, when detailed U.S. trade data are first available, and 2008, the most recent year of TRI reporting. By looking at changes to the TRI list, I identify the "treatment effect" of TRI regulation, which can be compared to net imports of a reference group of chemicals unaffected by a change to their TRI status. This allows us to differentiate a policy effect from a general rise in trade or other unobserved or idiosyncratic factors.

4. Data

The data used in this study come from U.S. Customs and Border Protection administrative records, which are processed by the Foreign Trade Division (FTD) of the U.S. Census Bureau. These transaction-level records include all merchandise exports and imports between the United States and its foreign trading partners over the period 1992 to 2008, comprising over 900 million unique commodity shipments.[27] I combine import and export transaction data to obtain values of net imports. Most export data information is compiled from Shipper's Export Declarations (SEDs), which American exporters are required to file with Customs officials at the port of export. Each SED contains information on a shipment of one or more types of merchandise from one American exporter to one foreign importer on a single transportation vessel.[28] As for imports, most of the data are collected via the U.S. Customs' Automated Commercial System, with the remainder from import entry summary forms, warehouse withdrawal forms, and Foreign Trade Zone forms.

Variables include information about the commodity, its transaction date, origin and destination, mode of transport, and processing. Among the characteristics describing the commodity are its Harmonized System (HS) classification code, current dollar value, quantities and units of measure, shipping weight, charges for tariff (in dollars), and transportation costs.[29] There are also derivative variables indicating whether it is high value or subject to a special tariff or trade program. Each transaction records the dates of exit and entry into the US, the merchandise's country of destination or origin, and domestic and foreign ports of entry and exit.[30] Also noted is how the merchandise is shipped (i.e., land, sea, or air), whether it was containerized, vessel manifest and visa numbers, and shipper identifiers. Furthermore, there are indicators for automated record processing, related party transactions, and use of imputed values.

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[27] The United States customs area includes all 50 states, the District of Columbia, and US possessions (e.g., Puerto Rico, Virgin Islands). These records do not include low value transactions (i.e., less than $2,501 for exports and less than $2,001 for imports), which are estimated using historical values per commodity per country.

[28] The exception to this rule is U.S.-Canadian trade, where the two countries have agreed to use each other's import data as the other's export data (i.e., data on U.S. exports to Canada come from Canadian authorities and are not filed by American exporters). This bilateral agreement does not include American exports in transit through Canada to another destination, which must be reported on an SED.

[29] Shipments to and from the U.S. are classified at the 10-digit level according to the U.S. Census Bureau's Schedule B commodity classification system (exports) and the Harmonized Tariff Schedule of the United States Annotated for Statistical Reporting Purposes (HTSUSA) (imports). For the same chemical, export and import codes may differ depending on national schedules as well as within each schedule over time. Exports are valued at their FAS (free alongside ship) value, which includes all transport and insurance charges incurred prior to foreign export. Imports are valued at their CIF (cost, insurance, and freight) value, which includes all costs incurred prior to arrival in the first American port of entry. In general, I will refer to both export and import schedules as HS schedules with clarification where appropriate.

[30] Canadian and Mexican transactions may also indicate foreign province in addition to port and country identifiers.

To identify traded HS commodities that are TRI chemicals, I mapped the list of chemicals that were added to the TRI by their chemical names. Of the 666 TRI chemicals, over 200 were added or dropped between 1992 and 2008; from these, I found exact name matches for fourteen chemicals with TRI listing changes in both the export and import schedules.[31] This number is limited primarily because most HS classifications, while highly detailed compared to industry classifications, nevertheless encompass multiple chemicals with at least one not on the TRI (or use a catchall category for commodities with small traded volumes). Furthermore, some exact matches in one schedule (e.g., imports) may not appear in the other, so I exclude these chemicals as well.[32]

The fourteen chemicals comprise seven that were added to the TRI list and seven that were removed between 1992 and 2008.[33] The former include alpha-hexachlorocyclohexane (added 1995), bromine (1995), chlorodifluoromethane (1994), fluorine (1995), formic acid (1994), lithium carbonate (1995) and vanadium oxides (2000).[34] The latter include acetone (dropped 1994), barium sulfate (1993), hydrochloric acid (1995), methyl ethyl ketone (2004), n-dioctyl phthalate (1993), phosphoric acid (1999), and sulfuric acid (1994).[35] Descriptive statistics of these chemicals are given in Table 1. Their uses range from household products (e.g., aerosols, cleaning fluids) to intermediates in (or byproducts of) industrial manufacturing (e.g., solvents, refrigerants), and also vary in their economic value, environmental hazard, and toxicity levels.[36] For example, chlorodifluoromethane (aka, HCFC-22) is the most widely used chemical in the panel, although hydrochloric acid has by far the most pounds of emissions in recent years; in contrast, formic acid is a chemical byproduct in the production of other chemicals like acetic acid and can be found naturally in some insect venom. Notwithstanding American production of all fourteen chemicals, which the U.S. has exported to 154 countries between 1992 and 2008, the U.S. also has imported them from 80 countries over the same period. These trade partners differ considerably in economic development, geography, resource endowment, and, by assumption, environmental protection.

5. Methodology

Variations in year of TRI listing or delisting, industrial production and usage, and countries of import origin and export destination can be credibly used in a differences-in-differences model to identify the impact of TRI

[31] For a full list of changes to the TRI list of toxic chemicals, see http://www.epa.gov/tri/chemical/list%20changes/ChemListChanges05.pdf.

[32] Note that besides differences between schedules, some codes have changed over time as well. These I account for through chemical name matches as opposed to the use of codes. To match these panel of TRI-listing changed chemicals whose those that did not experience a change, I used time-invariant commodity codes for the latter.

[33] Among these fourteen chemicals, three have multiple 10-digit commodity codes that are consistent over time. These include lithium carbonate (two codes), phosphoric acid (four), and vanadium oxide (two), which increases the panel size to nineteen discrete chemical entries. See data appendix for more detail.

[34] Alpha-hexachlorocyclohexane is more commonly known as alpha-lindane, and chlorodifluoromethane is known as HCFC-22. Vanadium oxides are included in the chemical category of vanadium compounds, which was added in 2000.

[35] Methyl ethyl ketone is also known as 1,3'-butadiene. Barium sulfate was removed from the listed category of barium compounds in 1993. The TRI listing for sulfuric acid was modified in 1994 to exclude non-aerosol forms; the chemical commodity code for sulfuric acid included in the treatment panel applies only to its liquid (oleum) form.

[36] See http://www.epa.gov/tri/chemical/hazardinfo/hazard_cx.htm for summary hazard information and http://www.scorecard.org/env-releases/def/tep_gen.html for a ranking of chemicals by toxicity level.

on domestic production via net imports. This methodology is commonly used to compare the difference in outcomes before and after a policy intervention between groups subject to the intervention and those that were not.[37] Assuming the exogeneity of the intervention and conditioning on time and treatment allow one to sidestep problems of endogeneity that are typical when comparing different individuals. For my study, this means comparing chemicals that were added to or removed from the TRI list (i.e., "treatment groups") with a variety of reference chemicals (i.e., "control groups"). These control groups include: 1) all other traded chemicals not included in the panel of chemicals whose TRI-listing changed; 2) the subset of all other chemicals that are traded each year; the subsets of all other chemicals either always or never listed in TRI that are traded each year; and 3) individual chemicals matched to each TRI-listing change chemical, but did not experience a listing change themselves (i.e., always or never listed in TRI). With the exception of the first reference group, the remaining control groups have time-consistent commodity codes for the entire period, thus mitigating bias from panel heterogeneity. As for the issue of policy exogeneity, nowhere in descriptions of the TRI program and regulation does the issue of international trade appear; moreover, given the fairly uniform composition of commercial chemicals (e.g., purity, quality), it seems reasonable to assume high substitutability of foreign for domestic production.[38]

To test the hypotheses that additions (deletions) to the TRI list correspond to increased (decreased) net imports of those chemicals affected, and that the sources (destinations) of subsequent new trade activity are countries with less (more) stringent environmental protection than the U.S., I estimate the following reduced form regression model with random effects for chemical, country, and year:

$$y_{ijt} = \beta_0 + \beta_{1i} x_{1i} + \beta_{2j} x_{2j} + \beta_{3t} x_{3t} + \beta_{4it} x_{4it} + \beta_{5ijt} x_{5ijt} + \varepsilon_{ijt}, \text{ where}$$

y_{ijt} = net import value for chemical i from country j in year t

x_{1i} = chemical i dummy variable

x_{2j} = country j dummy variable

x_{3t} = year t dummy variable

x_{4it} = TRI listing status dummy variable for chemical i in year t

x_{5ijt} = control variables for chemical i, country j, and year t

ε_{ijt} = error term

Since the dataset comprises all trade transactions for the U.S., I can construct a balanced panel for each chemical-country pair, including zero values for years in which trade for a chemical from a country does not occur. Each country's net trade with the U.S. for a given chemical is aggregated up to an annual value in current dollars, excluding transshipments (i.e., trade passing through a third country) and non-own production (i.e., chemicals are not produced in the country of export).

The main variable of interest is the policy intervention variable that varies by chemical and year, taking the value of zero for each year that the chemical is not listed on TRI and the value of one when listed. A positive coefficient can be interpreted as indicating a discrete increase in net import value due to TRI listing, which I argue is

[37] Some well-known examples using this methodology include Card and Krueger (1994) and Hastings (2004).
[38] Given their homogeneous nature, product differentiation within chemicals in order to price discriminate also seems unlikely.

analogous to a pollution haven effect. In addition to indicators for chemical, country, and year, I also include a number of annual variables by country that may contribute to trade flows separate from environmental policies.[39] First, there is country per capita income (GDP in current dollars divided by population), where a positive coefficient should correspond to the factor endowment hypothesis since chemical production is a capital-intensive activity, regardless whether policy intervention occurs. Moreover, when interacted with the TRI policy variable, this variable can provide evidence of the pollution haven hypothesis; that is, a negative coefficient indicates lower net imports of toxic chemicals from wealthier countries due to TRI listing.

With regard to gravity trade factors influencing trade patterns more generally, I include both a trade partner's overall GDP in current dollars as well as its bilateral distance to the U.S. measured between each country's largest city.[40] These may broadly indicate a trade partner's productive capacity and market demand as well as incurred trade transaction costs, respectively. Since the trade transaction data are highly detailed, however, I can more precisely identify the latter by also including both the actual cost of transport and freight for imports (current dollars) as well as the effective dutiable rate for each imported chemical (i.e., value of import duty divided by shipment value per year). Transport costs are expected to increase with net imports given a scaling effect, while tariff rates are expected to decrease net imports as they increase. Similarly, I include a country's terms of trade (i.e., buying power) with the U.S., which is the ratio of the prices earned from exports over those paid for imports and varies over time, to mitigate distortions from foreign exchange rate fluctuations and relative currency appreciation. This is because more favorable terms of trade (i.e., a ratio greater than one) are likely to correspond with increased net exports to the U.S.

Two additional variables are an indicator for OECD membership and the annual percent of related-party trade in imports for each chemical-country pair. Besides being convenient shorthand for wealthy economies, OECD membership may also proxy for institutional similarities and participation in selective multilateral treaties.[41] Thus, membership may further corroborate the pollution haven hypothesis if it corresponds to decreased net imports. Finally, I include the percent of related-party trade in imports for a given chemical since multinational activity may obscure trade patterns due to possible differences in pricing behavior.[42] Furthermore, it may also be the case that domestic production of toxic chemicals is disproportionately offshored to international subsidiaries.

6. Results

While the EPA reports that emissions for TRI chemicals have decreased in aggregate since their listing, the trends for each chemical in the panel of chemicals vary considerably. Table 1 provides a breakdown of each chemical's average annual emissions in the U.S. (column 3) and percent changes during their years of listing (columns 4 and 5). Among the group of chemicals that were delisted from TRI (lower panel), all exhibited lower

[39] Country-level variables, with the exception of bilateral distance, were obtained from the World Bank Group's World Development Indicators online database, whose permanent URL is: http://go.worldbank.org/3SGLDH5V10.
[40] Distances were calculated for the main city of each trading partner with New York City, using the online calculator: http://www.indo.com/distance/. One concern with using a single city per country is economic representativeness for large countries; in this paper, I include a direct measure of transport costs to mitigate this bias.
[41] A list of OECD member countries and their accession dates can be found at its website: http://www.oecd.org.
[42] See Tang (2010) and Bernard et al (2006).

levels of emissions from the time they were listed (or data collected) until their delisting, which may have accounted for their removal. For the seven chemicals that were added to TRI, however, three continue to show increased emissions despite being listed for over a decade. That said, a caveat to understanding these numbers is that the lack of emission data for these chemicals in the years when they were unlisted prevents pre- and post-policy comparison, so it is possible that the rate of emissions growth fell with regulation, somewhat achieving TRI's aims.[43]

[Table 1 here]

Fortunately, trade transaction data are available for all chemicals in both listed and unlisted years, and these values show that during the years of listing, half of the chemicals on average saw an increase in their net imports. In particular, five of the seven chemicals that were added to TRI experienced an average annual increase in their net import value during their listed years between 28 and 850 percent. Of these, three had an absolute decrease in emissions during the same period. As for the chemicals dropped from TRI, all seven decreased in overall emissions while only two had contemporaneously higher net imports. This trend appears to contradict the pollution haven effect since loosening environmental regulation may augur a decrease in their net imports as domestic firms expand production. However, it may also be the case that substitutes may have been found for these chemicals during their TRI-listed years, after which a lack of demand would inhibit greater production. Nevertheless, for the panel as a whole, the annual difference between unlisted and listed years is an increase in net import value exceeding $80 million, which, occurring alongside an average decrease of 69 million pounds of emissions during listed years, is consistent with the pollution haven effect.

The difficulty interpreting these annual averages in emissions and net imports, especially given their heterogeneous and contradictory trends, can be remedied with a richer model that also allows comparison with non-TRI chemicals and other reference groups. Results from the differences-in-differences regression analysis are presented in Tables 2 and 3. Table 2 uses all traded chemicals excluding those in the panel while Table 3 uses the subset of all other chemicals that are traded and retain the same commodity classification code for every year in the period (i.e., time-consistent HS codes). All figures (imports, duties, related party share) are aggregated up in annual value for treatment chemical-country and control group-country pairs.

[Table 2 here]

In Table 2, across all five specifications, the treatment indicator TRI listing (row 1) is positive and mostly significant statistically and economically.[44] Specifically, looking at the entire panel of treatment chemicals (columns A and B), inclusion in TRI is associated with an increase in a chemical's net import value ranging from

[43] Of course, the same argument can be made for chemicals that show a decrease during listed years, and that listing may have had no effect.

[44] An additional specification (not shown) includes only the TRI listing status change variable and country, chemical, and year fixed effects. I obtain a positive and significant coefficient on the TRI variable, consistent with the broader claim that regulatory change increases net imports (i.e., the pollution haven effect).

$10 to $14 million annually, which suggests a pollution haven effect. The increase is even more dramatic for the subset of TRI-added chemicals (column C), where net imports swell to $66 million on average per listed chemical per year, which is similar to the increase of $73 million in gross import value (column E). In contrast, the subset of TRI-dropped chemicals (column D) no longer exhibits a statistically significant increase in net imports for the TRI listing effect alone. As mentioned earlier, this may be due to changes in demand composition for listed chemicals (e.g., replacement with less toxic chemicals) and thus a muted effect from delisting.

More intriguing, however, is the interaction between TRI listing and per capita income (row 3), which has a statistically significant and negative coefficient for all specifications except that for TRI-dropped chemicals. The implication of this result is that the implementation of policy has led to the sourcing of imports from poorer countries. Assuming that per capita income corresponds to higher environmental protection, I interpret this finding as supporting the pollution haven hypothesis. For example, the results indicate that among countries whose per capita income levels are below $8,500, the value of their net exports of TRI-added chemicals (column C) to the U.S. will increase during listed years.[45] In contrast, with regard to TRI-dropped chemicals (column D), net imports will increase during listed years from all trade partners as indicated by the positive coefficient on the interaction term. At the same time, these results also corroborate the factor endowment hypothesis, as GDP per capita (row 2) on its own is positively and significantly associated with net imports across all specifications.

Other control variables in columns B through E further qualify American trade in these toxic chemicals relative to the rest of the chemical sector. Both GDP (row 4) and import transportation costs (row 5) are statistically significant with the expected signs in every specification, as gravity trade models would suggest.[46] Similarly, countries whose imports are subject to higher duty rates (row 6) or have more favorable terms of trade (row 7) export less of these chemicals in value, although this effect is statistically significant for the latter in gross terms (column E). Having accounted for per capita income variation, it is notable that OECD members (row 8) export substantially less of chemicals in value to the U.S. One interpretation of this result is that other characteristics associated with OECD membership, such as policy coordination or treaty adherence, mitigate the effect of capital intensive comparative advantage. Finally, the effect of related party trade share is strikingly different depending on whether the toxic chemicals in the panel are exported from or imported to the U.S. While imports from multinational firms (row 9) are higher in general compared to arms-length traders, during TRI listing years (row 10) this difference is markedly reduced (although still positive). Neither related party export share nor its interaction with TRI listing is statistically significant in any specification. Together, these suggest that while multinational companies may be predisposed to offshore toxic chemical production to their foreign affiliates and import them to the U.S., increased scrutiny on a firm's activities via TRI may extend across borders as well.

[Table 3 here]

[45] Trade partner per capita income thresholds, below which U.S. net imports of chemicals in the TRI-listing change panel will increase, are as follows: $14,212 (column B); $8,473 (column C); $6,546 (column E).
[46] Distance is included in all specifications, but not shown in the tables; given the presence of the directly measured transportation costs, it is unsurprising that its proxy bilateral distance is not significant in any specification.

Nevertheless, it may be the case that comparing the trade of all other chemicals to the panel of TRI chemicals whose listing status changed between 1992 and 2008 is biased since the composition of chemicals in the former may vary over time. Furthermore, one may be concerned that the panel of chemicals subject to regulatory change is unrepresentative of the TRI program as a whole. To ensure time consistency and representativeness of program scope, I separate from the all other chemicals group the subset of chemicals that are traded during the whole period and identify, based on exact chemical name matches, whether they were subject to TRI. This allows me to create two additional control groups, chemicals that were never on the TRI list and those that always were and compare these against the original panel of chemicals with a TRI listing change. These results are presented in Table 3 (columns A through C). In the specification using both control groups and the original panel (column A), neither TRI listing nor its interaction with GDP per capita is statistically significant. In comparing the panel with either each control group individually (columns B and C), however, I find that the interaction of TRI listing and per capita income is negative and statistically significant. In other words, while richer countries remain the primary exporters to the U.S. of the panel of toxic chemicals, following TRI listing they export relatively less. Coupled with the fact that net imports for the panel increase in total value, this means that poorer countries are exporting more of these chemicals during listing years in absolute and proportional terms.

Having identified individual chemicals that were either never or always on TRI, I can perform an additional robustness check using matched pairs with the panel of chemicals. I calculate the Mahalanobis distance between a chemical on the panel with each chemical in both the never and always TRI control groups, and match pairs based on probabilistic similarity in annual net import value and molecular composition (i.e., organic versus inorganic compounds).[47] Results from regressions using these matched pairs are found in columns D (chemicals never listed on TRI) and E (always TRI).[48] For chemicals matched with those never listed in TRI, TRI listing has no statistically significant effect on its own or interacted with GDP per capita; however, with always TRI matched pairs, both are significant and signed as hypothesized. That is, compared with similar reference chemicals that were always listed in TRI, those from the panel (added or dropped) experienced a discrete increase in net import value (row 1) during listing years, and were disproportionately sourced from poorer countries while listed (row 3), evidence supporting the pollution haven effect and pollution haven hypothesis, respectively.

7. Discussion

While declines in reported emissions of TRI chemicals in the years following their listing may indicate that TRI is having its intended regulatory effect, my results suggest that emissions may have simply relocated abroad. Using a panel of TRI chemicals whose listing status changed between 1992 and 2008, I identify a significant increase in net imports of chemicals that were listed on TRI relative to when they were not. Unlike earlier studies that analyze at the industry level, this study utilizes newly available trade transaction data that allow us to track net

[47] Mahalanobis distances identify similarities between variables based on proximity to the mean of their distributions, but unlike Euclidean distances also account for the covariance among variables; for example, see Bedrick et al (2000).

[48] Some chemicals in original TRI listing change panel were matched to the same chemical in the never or always TRI groups; this is reflected in the smaller number of matched control chemicals (row 13). Also, some matched chemicals in the control groups may have experienced a TRI listing change prior to 1992.

imports of specific chemicals affected by the TRI program. My estimates are sharpened not only through comparing pre- and post-TRI listing trade activity, but also by using the multiple reference groups and exploiting the variation between the TRI chemicals themselves, such as listing date, countries of origin and destination, and industrial use.

Besides failing to reject the pollution haven effect, where domestic production of toxic chemicals relocates abroad due to increased environmental regulation, I also find support for the pollution haven hypothesis, which suggests relocation to countries with less stringent protection policies. At the same time, my results are not inconsistent with the factor endowment hypothesis, which states that foreign sourcing of capital-intensive commodities will be from other wealthy countries regardless of their toxicity.

That said, these findings are contingent on a number of assumptions, in particular that per capita income is a good proxy for environmental protection. One extension of this research is to look at specific environmental regulations abroad that include the same chemicals covered under the TRI program, also known as pollution release and transfer registers (PRTRs). This may show more directly how variation in policies affects the trade patterns of the chemicals under regulation. One also might expect hypothesized effects to be stronger for chemicals that are more toxic and/or are released into the environment in greater quantities (and therefore are more noticeable to the public). A major problem, however, with this approach is that unlike trade value, which is usually high quality and comprehensive, quantity measurements vary depending on commodity, are subject to greater measurement error, and are less complete. Nevertheless, it may be possible to use a smaller sample of chemicals to study the direct toxicity impact from TRI listing.

Regression estimates for the variables of related party trade shares also suggest another area for further work. Whether these differences in import behavior are due to reputational or legal liability concerns that cross international borders may be clarified by examining more closely both the multinational and third party firms engaged in the chemicals trade. The trade transaction data can be linked to other firm-level data maintained by the U.S. Census Bureau, and knowing the nationality of a multinational's main office may be a separate dimension in which non-income differences (e.g., institutions) contribute to production offshoring decisions.

The results of this paper, however, are not contingent on the accuracy of reported emissions and complement existing studies about firm responses to the TRI program. Given possible threshold regarding or substitution to unlisted chemicals, domestic emissions of TRI chemicals (and their associated production and use) may not necessarily be declining despite reports otherwise. Nonetheless, a discrete increase in net imports coinciding with TRI listing indicates that domestic demand for listed toxic chemicals remains and that sourcing from abroad may be another means by which firms can comply with the letter of regulation if not its spirit.[49]

[49] See Bennear (2005) and Koehler and Spengler (2007) for extended discussions of possible firm responses to environmental regulation.

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Table 1: Chemicals with TRI-listing change, annual averages between 1988-2008

	Change year	Listed				Unlisted
		US emissions (mil lbs)	Annual emissions change (%)	Total emissions change (%)	Net imports* ($ mil)	Net imports* ($ mil)
Added chemicals						
alpha-hexachlorocyclohexane[1]	1995	n/a	n/a	n/a	1.30	4.16
bromine	1995	0.36	9.63	261.30	-2.80	-6.46
chlorodifluoromethane	1994	8.61	-1.92	-22.19	26.82	-15.57
fluorine	1995	0.14	12.56	413.67	-0.54	-0.75
formic acid	1994	11.17	1.63	23.40	-0.59	5.57
lithium carbonates	1995	0.27	-1.26	-69.71	10.51	-11.34
vanadium oxides[2]	2000	58.12	-8.46	-45.44	21.91	-2.92
Dropped chemicals						
acetone	1994	193.73	11.11	-35.94	-46.02	-115.56
barium sulfate[3]	1993	25.77	1.57	-2.27	7.95	8.83
hydrochloric acid	1995	344.37	-15.52	-54.20	-3.51	-2.68
methyl ethyl ketone	2004	78.58	-11.72	-80.67	-44.25	-4.54
n-dioctyl phthalate	1993	0.24	47.64	-5.00	-7.70	-0.94
phosphoric acids	1999	128.68	-8.43	-57.28	-144.89	-165.23
sulfuric acid, oleum[4]	1994	213.65	-5.89	-21.05	25.16	70.55

*Net import annual averages calculated for listed and unlisted periods between 1992 and 2008.
1: no emissions data were reported to the US EPA during listing period.
2: within vanadium compounds category. Emissions and percent changes are for the vanadium compounds category.
3: delisted from barium compounds category. Emissions and percent changes are for the barium compounds category.
4: delisted non-aerosol forms. Emissions and percent changes are for sulfuric acid prior to modification in 1993.
Source: US EPA, www.epa.gov/tri/; US Census Bureau.

Table 2: Differences-in-differences OLS regression results

DV: Net import value ($ mil)	[A] TRI change v. other chem.	[B] TRI change v. other chem.	[C] TRI additions v. other chem.	[D] TRI deletions v. other chem.	[E] TRI additions v. other chem. (gross value)
TRI listing	13.672***	10.176**	65.822***	3.138	73.003***
	(3.575)	(4.622)	(15.215)	(9.222)	(14.859)
GDP per capita ($1,000)	2.802***	4.801***	18.458***	8.083***	23.380***
	(0.252)	(0.386)	(1.081)	(0.624)	(1.056)
TRI listing · GDP p.c. ($1,000)	-0.960***	-0.716***	-7.768***	1.949***	-11.153***
	(0.196)	(0.256)	(0.665)	(0.540)	(0.648)
GDP ($100 bil)		-4.716***	-9.270***	-8.178***	-4.648***
		(0.829)	(1.726)	(1.384)	(1.688)
Import transport costs ($ mil)		10.048***	9.506***	10.100***	20.174***
		(0.385)	(0.575)	(0.489)	(0.555)
Import dutiable rate (%)		-5.067***	-6.749***	-7.729***	-8.748***
		(0.886)	(1.497)	(1.471)	(1.462)
Terms of trade (year 2000 = 100)		-0.042	-0.214	-0.060	-0.395**
		(0.088)	(0.181)	(0.127)	(0.177)
OECD member		-31.352**	-94.454***	-53.085**	-58.099*
		(14.199)	(31.024)	(22.307)	(30.339)
Related party import share (%)		0.592***	1.214***	0.707***	1.329***
		(0.096)	(0.218)	(0.126)	(0.213)
TRI listing · Related import (%)		-0.345**	-0.722**	-0.594**	-0.823***
		(0.156)	(0.317)	(0.285)	(0.310)
Related party export share (%)		-3.361	-7.999	0.586	9.073
		(9.341)	(24.505)	(13.005)	(23.933)
TRI listing · Related export (%)		8.043	16.652	-3.630	2.102
		(12.322)	(28.106)	(22.203)	(27.454)
Treatment / Control groups	19 / 1	19 / 1	9 / 1	10 / 1	9 / 1
Panel size	1,551	1,318	612	845	612
Observations	25,431	18,569	8,651	11,780	8,651
R-squared	0.049	0.082	0.154	0.106	0.459

Significance level: * 10 percent ** 5 percent *** 1 percent
All specifications include country, chemical, and year indicator variables. Panel identifier is by chemical (HS 10-digit level) and country, and standard errors are in parentheses.

Table 3: Robustness checks with time-consistent HS codes

DV: Net import value ($ mil)	[A] TRI change v. TRI controls	[B] TRI change v. never TRI	[C] TRI change v. always TRI	[D] TRI change v. never TRI (matched HS)	[E] TRI change v. always TRI (matched HS)
TRI listing	0.409	0.820	0.516	0.076	0.105**
	(0.976)	(0.862)	(0.515)	(0.061)	(0.048)
GDP per capita ($1,000)	0.461***	0.219***	0.266***	0.013***	0.009***
	(0.080)	(0.073)	(0.043)	(0.004)	(0.003)
TRI listing · GDP p.c. ($1,000)	-0.071	-0.078*	-0.056*	-0.003	-0.007**
	(0.054)	(0.048)	(0.029)	(0.003)	(0.003)
GDP ($100 bil)	-1.561***	-1.413***	0.334***	-0.024***	0.012*
	(0.171)	(0.154)	(0.092)	(0.009)	(0.007)
Import transport costs ($ mil)	16.404***	18.939***	4.184***	4.246***	3.511***
	(0.172)	(0.166)	(0.204)	(0.073)	(0.060)
Import dutiable rate (%)	-0.173	-0.106	-0.170	-0.022*	-0.014*
	(0.212)	(0.193)	(0.115)	(0.013)	(0.008)
Terms of trade (year 2000 = 100)	0.091***	0.088***	0.017*	0.004***	0.001
	(0.018)	(0.018)	(0.010)	(0.001)	(0.001)
OECD member	-17.107***	-13.599***	-5.581***	-1.342***	-0.602***
	(2.908)	(2.650)	(1.577)	(0.151)	(0.114)
Related party import share (%)	0.023	0.017	0.032***	0.013***	0.008***
	(0.022)	(0.022)	(0.011)	(0.001)	(0.001)
TRI listing · Related import (%)	0.033	0.0003	-0.010	-0.002	0.003*
	(0.030)	(0.028)	(0.018)	(0.002)	(0.002)
Related party export share (%)	-0.921	0.166	-1.705*	-0.322***	-0.278*
	(1.969)	(1.855)	(1.033)	(0.087)	(0.066)
TRI listing · Related export (%)	-1.130	-2.209	1.759	-0.012	-0.044
	(2.532)	(2.310)	(1.367)	(0.144)	(0.112)
Treatment / Control groups	19 / 2	19 / 1	19 / 1	19 / 17	19 / 18
Panel size	1,450	1,312	1,317	2,018	2,192
Observations	20,335	18,489	18,553	28,856	31,337
R-squared	0.198	0.296	0.060	0.259	0.255

Significance level: * 10 percent ** 5 percent *** 1 percent
All specifications include country, chemical, and year indicator variables. Panel identifier is by chemical (HS 10-digit level) and country, and standard errors are in parentheses.

Appendix: Panel of TRI chemicals with listing changes, 1992-2008

Description	Health hazard	Major markets, 2008 (import/export)	TRI chemical matches, 1992 (always/never listed)*
acetone **CAS 67-64-1** Clear liquid with sweet odor. Used to produce chemicals and solvents.	Moderate (NFPA Level 2) Skin, eye, nose, and throat irritant. Can cause headache, vertigo, nausea, vomiting, and unconsciousness.	South Africa, Spain / China, Mexico	mixed xylene isomers / aminoglycoside antibiotics
alpha-hexachlorocyclohexane (aka, alpha-lindane) **CAS 319-34-6** White crystalline powder. Used as insecticide and is a byproduct of lindane production.	Moderate (NFPA Level 2) Skin and eye irritant. Can cause coughing, wheezing, shortness of breath, nausea, convulsions, level and kidney failure, and cancer.	India / Germany, Italy	tetrachloroethylene / coumarin, methylcoumarin, ethylcouramin
barium sulfate **CAS 7727-43-7** White crystalline powder. Used to produce chemicals and pigments.	Minimal (NFPA Level 0)	China, Germany / Canada, Mexico	lead monoxide / calcium chloride
bromine **CAS 7726-95-6** Dark red volatile liquid. Used in fire retardant, dyes, drugs, fumigants, gasoline additive, and water treatment.	Serious (NFPA Level 3) Skin, eye, nose, and throat irritant. Can cause coughing, shortness of breath, and pulmonary edema.	Israel, Mexico / Canada, United Arab Emirates	silver nitrate / phosphorus sulfide
chlorodifluoromethane (aka, HCFC-22) **CAS 75-45-6** Clear gas with sweet odor. Used as refrigerant and solvent.	Serious (NFPA Level 3) Skin, eye, nose, and throat irritant. Can cause shortness of breath, pulmonary inflammation, chemical burn, and respiratory failure.	China, Germany / Mexico, Puerto Rico	cresols and their salts / cyclohexanone

Description	Health hazard	Major markets, 2008 (import/export)	TRI chemical matches, 1992 (always/never listed)*
fluorine **CAS 7782-41-4** Yellowish green gas with sharp odor. Used to produce chemicals and rocket fuel.	Extreme (NFPA Level 4) Skin, eye, nose, and throat irritant. Can cause shortness of breath, vomiting, pulmonary edema, osteosclerosis, and liver and kidney damage.	South Africa / India, South Korea	copper chloride oxide, hydroxide / chlorites, hypochlorites, hypobromites
formic acid **CAS 64-18-6** Clear liquid with strong odor. Used to produce textiles, paper, leather, fumigant, pesticide, solvents, glass, and in electroplating.	Serious (NFPA Level 3) Skin, eye, nose, and throat irritant. Can cause nausea, headache, vertigo, shortness of breath, and kidney damage.	China, Germany / Canada, Mexico	tetrachloroethylene / benzyl alcohol
hydrochloric acid **CAS 7647-01-0** Clear gas with strong odor. Used to produce chemicals and metals and as disinfectant.	Serious (NFPA Level 3) Skin, eye, nose, and throat irritant. Can cause bronchitis, pulmonary edema, and liver and kidney damage.	Canada, Mexico / Canada, Germany	phosphorus oxychloride, trichloride / silicon
lithium carbonate **CAS 554-13-2** White granular powder. Used to produce aluminum, glass, ceramics, and drugs, and as chemical catalyst.	Moderate (NFPA Level 2) Eye irritant. Can cause tremors, vertigo, dehydration, weight loss, and kidney and thyroid damage.	Argentina, Chile / Canada, Japan	silver nitrate; chromium trioxide / double, complex silicates; calcium carbide
methyl ethyl ketone (aka, 1,3'-butadiene) **CAS 78-93-3** Clear liquid with menthol odor. Used to produce plastics, textiles, paint, and solvents.	Slight (NFPA Level 1) Skin, eye, and digestive irritant. Can cause headache, nausea, numbness, unconsciousness, coma, and respiratory failure.	South Africa, United Kingdom / Canada, Mexico	o-xylene / vanillin

Description	Health hazard	Major markets, 2008 (import/export)	TRI chemical matches, 1992 (always/never listed)*
n-dioctyl phthalate **CAS 117-84-0** Clear liquid with mild odor. Used to produce plastics, coatings, and adhesives.	Minimal (NFPA Level 0) Skin and eye irritant. Can cause shortness of breath and wheezing.	Brazil, South Korea / Canada, Mexico	chloroform / furfuryl alcohol, tetrahydrofurfuryl alcohol
phosphoric acid **CAS 7664-38-2** Clear, odorless solid or liquid. Used in fertilizers, detergents, food products, and water treatment.	Moderate (NFPA Level 2) Skin, eye, nose, and throat irritant. Can cause dry skin, coughing, bronchitis, and shortness of breath.	Belgium, Israel / India, Mexico	copper chloride oxide, hydroxide; thorium compounds; sodium cyanide, cyanide oxide; mercury / phosphorus pentachloride, uranium metal; miscellaneous sodium sulfates
sulfuric acid **CAS 7664-93-9** Clear or brownish, odorless liquid. Used to produce batteries, fertilizers, paper, textiles, explosives, ferrous metals, and drugs.	Serious (NFPA Level 3) Skin, eye, nose, and throat irritant. Can cause headache, nausea, vomiting, pulmonary edema, and cancer.	Canada, India / Canada, Venezuela	manganese dioxide / silicon
vanadium oxide **CAS 1314-34-7 and 1314-62-1** Yellow or red crystalline powder. Used to produce dyes and chemicals and as photographic developer.	Serious (NFPA Level 3) Skin, eye, nose, and throat irritant. Can cause shortness of breath, coughing, and greenness of tongue.	China, Russia / Canada, Russia	arsenic; phosphorus oxychloride, trichloride / sodium hexafluoroaluminate; miscellaneous sodium silicates

*: Per Table 3, columns D and E, TRI listing-change chemicals with multiple HS codes are individually matched by HS and other Mahalanobis criteria; see text.
Source: New Jersey Department of Health and Senior Services, website: http://web/doh.state.nj.us/rtkhsfs/search.aspx.